

Binary Classification of Seizures in EEG Data

STATGR5243_001_2025_3 – Applied Data Science

Bryndis Lif Bjarnadottir

Department of Biomedical Engineering
Columbia University
New York, NY, USA
blb2181@columbia.edu

Elias Ludviksson

Department of Mechanical Engineering
Columbia University
New York, NY, USA
el3403@columbia.edu

ABSTRACT

Roughly 1–2% of the global population suffers from epilepsy, a disease that causes seizures which affect brain function and can impair consciousness or movement. Here, four machine learning algorithms are applied to the CHB-MIT Scalp EEG database for binary seizure classification. A pre-processed version of the dataset is utilized, in which the continuous EEG data has been segmented into 8-second epochs. Features are hand-crafted and include time-, frequency- and nonlinear features. To mitigate the class imbalance caused by the predominance of non-seizure epochs, the latter are undersampled. Logistic regression achieved the lowest overall performance with a precision of 0.719, recall of 0.357 and F1-score of 0.477. Then came Random Forest with a precision of 0.813, recall of 0.641 and F1-score of 0.717. Support Vector Machines reported a precision of 0.778, recall of 0.690 and F1-score of 0.732. The best performing model was Gradient Boosting with a precision of 0.798, recall of 0.744 and F1-score of 0.770. These findings highlight Gradient Boosting as the most promising approach among the tested algorithms for reliable seizure classification, while also indicating that performance remains limited.

I. INTRODUCTION

Approximately 1-2% of the world population suffers from epilepsy, a neurological disorder that is particularly prevalent among children, with an estimated 4% affected worldwide [1]. Epilepsy is defined by transient seizures which arise from abnormal brain activity and can lead to problems with attention, memory, speech, execution and the ability to plan and carry out coordinated, purposeful movements [2].

Epilepsy is divided into generalized and focal seizures. In generalized epilepsy, seizure activity begins in both brain hemispheres at once, affecting the whole body and often causing loss of consciousness, stiffness or jerking movements. In focal epilepsy, abnormal activity starts in one hemisphere and may cause movements or changes in awareness on one side of the body. It can sometimes spread to the other hemisphere through secondary generalization [1].

Electroencephalography (EEG) is a common and non-invasive way to detect seizures [3]. However, EEG recordings are often lengthy and complex, making their interpretation

time-consuming and dependent on expert analysis [4]. The signals are also noisy and seizures can look different depending on the type of epilepsy the patient has [5]. To address this challenge, machine learning techniques have been developed to automatically detect seizures from EEG data, as will be discussed in Section II.

This study uses the Children’s Hospital Boston-MIT (CHB-MIT) Scalp EEG Database, a publicly available dataset containing EEG recordings from pediatric patients with epilepsy. The dataset includes both focal and generalized seizures, giving a range of brain activity patterns for training and testing. The input to our algorithm is EEG signal data, and the model outputs a prediction of whether a seizure is occurring during a given epoch. The goal of this study is to compare the performance of Logistic Regression (LR), Support Vector Machine (SVM), Random Forest (RF) and Gradient Boosting (GB) models in a binary classification of seizures.

II. RELATED WORK

LR provides a simple and well-defined algorithm for binary classification, making it a good baseline model. Early seizure detection studies often used LR with basic hand-crafted features. Their results were modest, with [6] reporting a sensitivity of 88%. More recent work reported accuracy above 99%, but these results were produced using the Bonn dataset [7], which is a clean, balanced and a single-channel EEG dataset. This makes the classification task considerably easier than working with CHB-MIT, where seizures are rare, the signals are noisy and recordings span many patients. Spline-based extensions of LR can in theory capture nonlinear relationships in EEG data. However, spline-enhanced LR is rarely used as a classifier in seizure detection. Instead, splines are mostly applied in feature extraction [8] or preprocessing [9]. For these reasons, and because the focus of this study is on comparing standard classical models, spline-based LR is discussed here for completeness but not included among the implemented methods.

SVM has an established history in seizure detection. In [10], they showed that SVM can achieve strong patient-dependent seizure classification performance (an accuracy of 0.9687, sensitivity of 0.7299 and specificity of 0.9813), whereas patient-independent performance was noticeably lower. Because SVM

performance is highly sensitive to its hyperparameters, a systematic tuning strategy is essential. This is precisely what [11] implemented. They swept a range of C values across 2^j for $j = -4, -3, \dots, 4$ and simultaneously tuned the RBF kernel parameter γ over the same range, using 5-fold cross-validation to select the optimal configuration (although the specific best values were not reported). Using the tuned SVM to classify seizures in EEG data, their model achieved an accuracy of 0.72, a precision of 0.7398 and an F1-score of 0.8228.

RF is also commonly used in seizure detection on EEG data. In [12], RF was applied to EEG windows and achieved an accuracy of 0.9572, a precision of 0.9540 and a recall of 0.8197. However, these results were obtained on the Bonn dataset. RF has also been applied in settings that involve hand-crafted EEG features and hyperparameter tuning. For example, [13] used a grid-searched RF model for EEG classification, illustrating that RF can benefit from systematic parameter optimization, similar to the approach used in this study.

In [11], a GB decision tree classifier was also implemented and outperformed the SVM model. The authors applied 5-fold cross-validation for hyperparameter selection, but did not report which hyperparameters were considered.

III. DATASET

A. Dataset Description

This project uses EEG data from the public CHB-MIT Scalp EEG database, which contains long-term scalp EEG recordings from pediatric patients with epilepsy [14]. The original recordings are multi-channel time series sampled at 256 Hz with seizure annotations provided by clinical experts.

This work uses a preprocessed dataset, sourced from Kaggle, in the form of fixed-length EEG segments [15]. The raw signal data is stored in `signal_samples.npy` with shape (9505, 18, 2048). Each sample is an 8-second window (2048 time points at 256 Hz) recorded from 18 EEG channels. The corresponding label file `is_sz.npy` has shape (9505,) and contains, for each window, the fraction of time points that are annotated as seizure in the original CHB-MIT labels.

To obtain a binary classification target, each window is labeled as seizure if any part of the 8-second segment contains seizure activity, and non-seizure otherwise. With this definition, the dataset contains 9505 windows in total, of which 2581 are seizure windows and 6924 are non-seizure windows. Hand-crafted features are computed for each 8-second window and used as inputs to the classification models, as will be discussed in further detail below.

B. Preprocessing

To reduce class imbalance, the non-seizure windows are randomly downsampled to obtain a target ratio of approximately 1:2 (seizure : non-seizure), following the preprocessing strategy presented in [16]. This procedure yields a dataset of 7743 windows. Missing values in the engineered features (see next subsection) are handled using mean imputation via `SimpleImputer`. For models sensitive to feature scale, such

as LR, SVM and GB, feature standardization is applied using `StandardScaler`. The scaler is fit using only the training data and subsequently applied to the validation and test sets to avoid information leakage. RF is trained on the unscaled features.

C. Feature Extraction

Each 8-second EEG window is transformed into a feature vector composed of 12 hand-crafted frequency features, 8 time features and 3 nonlinear features, as done in [16]. For every time window the following features are computed.

- **Time-domain features:** Maximum, minimum, mean, median, standard deviation, variance, kurtosis, skewness, number of zero-crossings, proportion of positive samples, max/min ratio and mean absolute value.
- **Frequency-domain features:** Features derived from the power spectral density (PSD) estimated using Welch's method, including maximum, minimum, mean, median, standard deviation, variance, kurtosis and skewness of the PSD.
- **Nonlinear features:** Sample entropy, permutation entropy and Katz fractal dimension.

This results in 23 features per channel. To obtain a single representation for each window, the channel-wise feature vectors are averaged across the 18 EEG channels. The final feature matrix therefore has shape (7743, 23), where the rows represent EEG windows and the columns represent engineered features.

D. Data Partitioning

After preprocessing and feature extraction, the dataset is divided into training, validation and test sets using a patient-dependent split, as shown to be effective in [10]. The final split follows a 70/15/15 distribution. The validation set is used exclusively for hyperparameter tuning and the test set is reserved for the final evaluation of all models.

IV. METHODS

A. Problem Formulation

Given a feature vector $x \in \mathbb{R}^d$ derived from an 8-second EEG epoch, the objective is to predict a corresponding class label $y \in \{0, 1\}$. We define the positive class ($y = 1$) as an epoch containing seizure activity and the negative class ($y = 0$) as a non-seizure epoch. The goal of the learning algorithm is to approximate the function $f : x \rightarrow y$ that minimizes the classification error on unseen data.

B. Preliminary Data Processing Approach

Prior to adopting the pre-processed dataset described in Section III, we attempted an end-to-end pipeline starting from the raw CHB-MIT `.edf` files. In this initial phase, we implemented a custom preprocessing routine using the MNE-Python library. We standardized the electrode montage to 18 channels and segmented the continuous recordings into 2-second non-overlapping epochs, following the methodology outlined in [17].

This preliminary dataset was highly imbalanced, containing over 1.7 million non-seizure samples against fewer than 6,000 seizure samples (a prevalence of $< 0.4\%$). The sheer volume of this data made the pipeline computationally expensive and significantly prolonged training times. We experimented with LR, RF, GB (XGBoost) and CNN models on different training sets. However, the models struggled to generalize effectively due to the extreme imbalance. For instance, while we achieved a maximum recall of 92%, the corresponding precision was only 28%. The highest precision achieved was 47% with 79% recall using XGBoost, while the CNN model also yielded similar results with 46% precision and 84% recall on the same training set. These consistently low precision scores indicated an unmanageable false positive rate. Consequently, driven by the need to reduce computational costs and address the extreme class imbalance, we pivoted to the curated dataset [15], which utilizes longer 8-second windows and a balanced class distribution, allowing for more robust feature extraction and stable model performance.

C. Logistic Regression

As discussed in [18], LR models the probability of a binary outcome using the logistic function. For features $x \in \mathbb{R}^p$ and parameters (β_0, β) , the model assumes:

$$\log\left(\frac{\Pr(Y = 1 \mid X = x)}{\Pr(Y = 0 \mid X = x)}\right) = \beta_0 + x^\top \beta, \quad (1)$$

which corresponds to the familiar probability form:

$$\Pr(Y = 1 \mid X = x) = \frac{1}{1 + \exp(-(\beta_0 + x^\top \beta))}. \quad (2)$$

In practice, LR is typically fitted using a regularized maximum-likelihood criterion. Software implementations such as `scikit-learn` introduce an ℓ_2 penalty $\frac{1}{2C} \|\beta\|_2^2$, where C controls the strength of regularization (smaller C applies a strong penalty, while a large C does the opposite). In this study, the effect of the regularization parameter C was examined by evaluating models with $C \in \{0.01, 0.1, 1, 10\}$ and selecting the configuration that achieved the highest F1-score.

D. Support Vector Machine

SVM constructs a hyperplane in a high-dimensional feature space to separate classes with the maximal margin [18]. We utilized the nonlinear Radial Basis Function (RBF) kernel to handle complex, nonlinear relationships in the EEG data. The optimization objective minimizes the squared norm of the weight vector $\mathbf{w}$ while penalizing misclassifications through slack variables ξ_i :

$$\min_{\mathbf{w}, b, \xi} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^n \xi_i \quad (3)$$

subject to $y_i(\mathbf{w}^T \phi(x_i) + b) \geq 1 - \xi_i$ and $\xi_i \geq 0$ [19]. Here, the parameter C (regularization) controls the trade-off between

maximizing the margin and minimizing classification errors on the training data. The RBF kernel function is defined as:

$$K(x, x') = \exp(-\gamma \|x - x'\|^2) \quad (4)$$

where γ defines the influence of a single training example. In this study, we performed a grid search to optimize these hyperparameters, evaluating $C \in \{0.1, 1.0, 10.0\}$ and $\gamma \in \{\text{'scale'}, 0.01, 0.1\}$.

E. Random Forest

RF is an ensemble learning method that operates by constructing a multitude of decision trees at training time, a technique known as bootstrap aggregating or "bagging" [20]. The final classification is obtained by a majority vote of the individual trees, which helps reduce variance and mitigate overfitting compared to single decision trees. The mathematical structure relies on the aggregation of B trees, where the prediction $\hat{f}(x)$ for a new point x is:

$$\hat{f}(x) = \text{mode}\{\hat{C}_b(x)\}_{b=1}^B \quad (5)$$

where $\hat{C}_b(x)$ is the class prediction of the b -th tree. In our implementation, the model capacity was tuned via three key hyperparameters: `n_estimators` (B), which sets the number of trees in the forest; `max_depth`, which limits the maximum depth of the tree to prevent overfitting; and `min_samples_leaf`, which guarantees a minimum number of samples in a terminal node. We evaluated configurations with `n_estimators` $\in \{100, 300\}$, `max_depth` $\in \{\text{None}, 10, 20\}$, and `min_samples_leaf` $\in \{1, 2\}$.

F. Gradient Boosting

GB builds an additive model in a forward stage-wise fashion, where each new tree is trained to predict the pseudo-residuals (errors) of the preceding ensemble [21]. Unlike RF, which builds independent trees, GB constructs trees sequentially. The model update at step m is given by:

$$F_m(x) = F_{m-1}(x) + \nu \cdot h_m(x; \theta_m) \quad (6)$$

where h_m is the weak learner (decision tree) parameterized by θ_m and ν ($0 < \nu \leq 1$) is the `learning_rate` parameter that scales the contribution of each tree to prevent overfitting [18]. The complexity is further controlled by `n_estimators` (the total number of boosting stages M) and `max_depth` (the interaction depth of the trees). We also employed stochastic gradient boosting parameters: `subsample`, which is the fraction of samples used for fitting the individual base learners, and `max_features`, the number of features to consider when looking for the best split. An extensive grid search was conducted over `learning_rate` $\in \{0.05, 0.1\}$, `n_estimators` $\in \{1000, 1500, 2000\}$, `max_depth` $\in \{3, 4, 5\}$, `subsample` $\in \{0.8, 1.0\}$ and `max_features` $\in \{\text{'sqrt'}, \text{None}\}$.

V. EXPERIMENTS, RESULTS AND DISCUSSION

A. Logistic Regression

Four values of the hyperparameter C were evaluated and $C = 10$ achieved the highest F1-score. However, its performance advantage over the other settings was small. Table I shows the test results for LR, which noticeably performed the worst among all models. The model reached an accuracy of 0.739 but its recall was much lower at 0.357, indicating that many seizure epochs were missed. In contrast, its precision of 0.719 shows that the seizure predictions it did make were often correct. This imbalance between precision and recall results in a low F1-score of 0.477, demonstrating that LR struggles to capture the patterns necessary for reliable seizure detection. Figure 1 shows the corresponding learning curve, which helps interpret the test results. The validation F1 remains nearly flat despite increasing training size indicating underfitting and explaining why the model fails to generalize well on the test set.

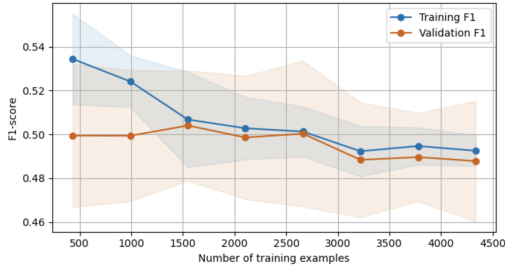


Fig. 1: Learning curve of the best LR model. Training F1 decreases as more data is used, while validation F1 slowly improves and the gap between them narrows, indicating reduced overfitting as the dataset grows. The shaded regions show ± 1 standard deviation across the cross-validation folds.

B. Support Vector Machine

Of the nine combinations of C and γ tested for the SVM, the configuration with $C = 10$ and $\gamma = 0.1$ achieved the highest validation F1-score (Figure 2). Using these parameters in the testing phase, the RBF SVM reached a test accuracy of 0.831 and an F1-score of 0.732, with a recall of 0.690 and a precision of 0.778. These results show that the model was able to detect a meaningful proportion of seizure epochs (267 of 387) while maintaining a moderate rate of false positives. Overall, the RBF SVM's results are comparable to those of RF and GB and markedly better than LR, particularly in terms of recall. Its AUC of 0.893 further indicates its ability to separate seizure from non-seizure epochs. Although polynomial and sigmoid kernels were also evaluated, they performed substantially worse, reinforcing that the RBF kernel is the most suitable choice for this task.

C. Random Forest

The results from the hyperparameter grid search show that the validation F1-scores vary only slightly across the tested configurations (0.683–0.734), suggesting that the RF model

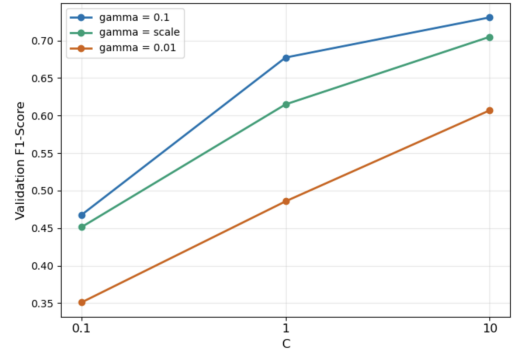


Fig. 2: Validation F1-score of the SVM classifier across the tested combinations of the regularization parameter $C \in \{0.1, 1, 10\}$ and kernel parameter $\gamma \in \{0.01, 0.1, \text{scale}\}$. The highest F1-score was obtained with $C = 10$ and $\gamma = 0.1$.

is fairly robust to changes in these hyperparameters within the explored range. The best validation performance was achieved with `n_estimators=100`, `max_depth=None`, and `min_samples_leaf=1`, indicating that allowing deeper trees was beneficial. When evaluated on the test set, this configuration yielded a precision of 0.813, an accuracy of 0.831, a recall of 0.641, an F1-score of 0.717 and an AUC of 0.913. These results show that the RF model is capable of identifying a substantial proportion of seizure epochs (284 out of 387) while producing relatively few false positives (57 non-seizure epochs misclassified as seizures out of 775).

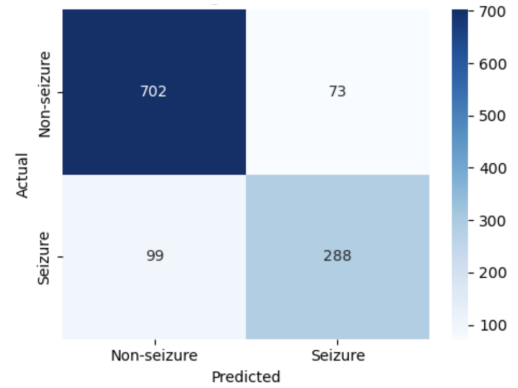


Fig. 3: The GB model shows high true negatives (702) and true positives (288), with 73 false positives and 99 false negatives.

D. Gradient Boosting

Out of the 72 GB configurations, the best validation F1-score was obtained with `learning_rate=0.1`, `n_estimators=1500`, `max_depth=4`, `subsample=1.0`, and `max_features='sqrt'`. When evaluated on the test set, this model achieved the strongest performance among the classical approaches (Table I). It obtained an accuracy of 0.852 and a recall of 0.744, which was the highest among the models considered and indicates that the GB classifier was able to detect a relatively large proportion of seizure epochs. However,

TABLE I: Test performance of all evaluated models. The table reports accuracy, precision, recall, F1-score and area under the ROC curve (AUC) on the held-out test set.

Model	Accuracy	Precision	Recall	F1-Score	AUC
LR	0.739	0.719	0.357	0.477	0.766
SVM	0.831	0.778	0.690	0.732	0.893
RF	0.831	0.813	0.641	0.717	0.913
GB	0.852	0.798	0.744	0.770	0.919

the model still missed 99 out of 387 seizure events, showing that false negatives remain a concern in this setting.

The model also produced a precision of 0.798 and an overall F1-score of 0.770, reflecting a balanced compromise between detecting seizures and avoiding false alarms. The confusion matrix in Figure 3 illustrates this balance, the model yielded 288 true positives and 702 true negatives, alongside 73 false positives and 99 false negatives. Overall, the classifier demonstrates good discriminative ability between seizure and non-seizure epochs, though improvements in recall would be important for clinical use.

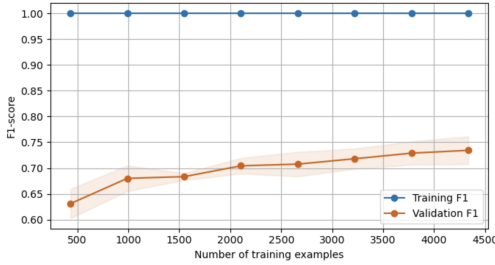


Fig. 4: Learning curve of the best GB model. The training F1 remains 1.0 across all training sizes, while the validation F1 increases gradually and stays notably lower, indicating overfitting. The shaded regions show ± 1 standard deviation across the cross-validation folds.

Figure 4 shows the learning curve for the GB model. The training F1-score remains 1.0 across training sizes, while the validation F1-score increases steadily but remains substantially lower. This gap indicates overfitting, the model fits the training data extremely well but does not generalize equally strongly to unseen samples. The curve also shows that adding more training data improves validation performance only gradually suggesting that architectural or regularization adjustments may be more effective than simply increasing data volume.

VI. CONCLUSION AND FUTURE WORK

Among the four models implemented, GB achieved the best performance with an F1-score of 0.770, accuracy of 0.852 and recall of 0.744. SVM and RF produced moderate results, while LR performed the worst. Our findings indicate that although classical machine learning with hand-crafted features can achieve moderate success, the false negative rate remains a challenge for clinical deployment. Future work would benefit from exploring approaches that classify directly from the raw EEG signal rather than relying on hand-crafted epochs.

Additionally, it would be interesting to investigate whether different types of seizures can be reliably classified from the EEG signal.

CONTRIBUTIONS

B.L.B. and E.L. jointly contributed to the machine learning methodology. Both authors explored and evaluated multiple algorithms, implemented the associated code and participated in the development and refinement of the analytical pipeline.

ACKNOWLEDGMENT

The authors would like to thank Professor Bianca Dumitrascu and Teaching Assistant Jonathan Huml for their guidance and feedback throughout the course. The authors also acknowledge ChatGPT for assistance with wording, proofreading and code implementation.

REFERENCES

- [1] N. S. Voros and C. P. Antonopoulos, Eds., *Cyberphysical Systems for Epilepsy and Related Brain Disorders: Multi-parametric Monitoring and Analysis for Diagnosis and Optimal Disease Management*. Switzerland: Springer International Publishing, 2015. [Online]. Available: <https://doi.org/10.1007/978-3-319-20049-1>
- [2] R. S. Fisher, W. van Emde Boas, W. Blume, C. Elger, P. Genton, P. Lee, and J. E. Jr., "Epileptic seizures and epilepsy: definitions proposed by the international league against epilepsy (ilae) and the international bureau for epilepsy (ibe)," *Epilepsia*, vol. 46, no. 4, pp. 470–472, 2005. [Online]. Available: <https://doi.org/10.1111/j.0013-9580.2005.66104.x>
- [3] J. M. Stern, J. E. Jr., and J. Engel, *Atlas of EEG Patterns*. Philadelphia, PA: Wolters Kluwer, 2013. [Online]. Available: <https://ebookcentral.proquest.com/lib/columbia/detail.action?docID=3418257>
- [4] S. Wong, A. Simmons, J. Rivera-Villicana, S. Barnett, S. Sivathamboo, P. Perucca, P. Kwan, L. Kuhlmann, R. Vasa, and T. J. O'Brien, "Eeg based automated seizure detection – a survey of medical professionals," *Epilepsy & Behavior*, vol. 149, p. 109518, 2023. [Online]. Available: <https://doi.org/10.1016/j.yebeh.2023.109518>
- [5] R. K. Maganti and P. Rutecki, "Eeg and epilepsy monitoring," *Continuum (Minneapolis)*, vol. 19, no. 3, pp. 598–622, 2013. [Online]. Available: <https://doi.org/10.1212/01.CON.0000431378.51935.d8>
- [6] A. Alkan, E. Koklukaya, and A. Subasi, "Automatic seizure detection in eeg using logistic regression and artificial neural network," *Journal of Neuroscience Methods*, vol. 148, no. 2, pp. 167–176, 2005. [Online]. Available: <https://doi.org/10.1016/j.jneumeth.2005.04.009>
- [7] S. Zeini, S. E. Alavi, and K. A. Asl, "Using nonlinear features and logistic regression for epilepsy detection with linear complexity," *Frontiers in Health Informatics*, vol. 13, p. 183, 2024. [Online]. Available: <https://doi.org/10.30699/fhi.v13i0.535>
- [8] Z. R. Zamir, "Detection of epileptic seizure in eeg signals using linear least squares preprocessing," *Computer Methods and Programs in Biomedicine*, vol. 133, pp. 95–109, 2016. [Online]. Available: <https://doi.org/10.1016/j.cmpb.2016.05.002>
- [9] J. Kayser and C. E. Tenke, "Issues and considerations for using the scalp surface laplacian in eeg/erp research: A tutorial review," *International Journal of Psychophysiology*, vol. 97, no. 3, pp. 189–209, 2015. [Online]. Available: <https://doi.org/10.1016/j.ijpsycho.2015.04.012>
- [10] S. Janjarsjitt, "Epileptic seizure classifications of single-channel scalp eeg data using wavelet-based features and svm," *Medical and Biological Engineering and Computing*, vol. 55, no. 10, pp. 1743–1761, 2017. [Online]. Available: <https://doi.org/10.1007/s11517-017-1613-2>
- [11] J. He, L. Yang, D. Liu, and Z. Song, "Automatic recognition of high-density epileptic eeg using support vector machine and gradient-boosting decision tree," *Brain Sciences*, vol. 12, p. 1197, 2022. [Online]. Available: <https://doi.org/10.3390/brainsci12091197>
- [12] A. Gokturk and J. Luu, "Epileptic seizure detection using machine learning on electroencephalogram data," *Journal of Emerging Investigators*, vol. 7, no. 1, pp. 1–6, 2024. [Online]. Available: <https://doi.org/10.59720/24-028>

- [13] X. Wang, G. Gong, N. Li, and S. Qiu, "Detection analysis of epileptic eeg using a novel random forest model combined with grid search optimization," *Frontiers in Human Neuroscience*, vol. 13, p. 52, 2019. [Online]. Available: <https://doi.org/10.3389/fnhum.2019.00052>
- [14] J. Gutta, "Chb-mit scalp eeg database (version 1.0.0)," 2010, rRID:SCR_007345. [Online]. Available: <https://doi.org/10.13026/C2K01R>
- [15] M. Gotoh, "Mit-chb processed," Kaggle Online Notebook, 2021, accessed: 2025-12-05. [Online]. Available: <https://www.kaggle.com/datasets/masahirogotosh/mit-chb-processed>
- [16] Ava225, "15 channels mit-chb," Kaggle Online Notebook, 2025, accessed: 2025-12-05. [Online]. Available: <https://www.kaggle.com/code/ava225/15-channels-mit-chb>
- [17] A. H. Shoeb, "Application of machine learning to epileptic seizure onset detection and treatment," Ph.D. dissertation, Massachusetts Institute of Technology, Cambridge, MA, 2009. [Online]. Available: <http://hdl.handle.net/1721.1/54669>
- [18] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, 2nd ed. New York, NY: Springer, 2009. [Online]. Available: <https://doi.org/10.1007/978-0-387-84858-7>
- [19] C. Cortes and V. Vapnik, "Support-vector networks," *Machine Learning*, vol. 20, no. 3, pp. 273–297, 1995. [Online]. Available: <https://doi.org/10.1007/BF00994018>
- [20] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001. [Online]. Available: <https://doi.org/10.1023/A:1010933404324>
- [21] J. H. Friedman, "Greedy function approximation: A gradient boosting machine," *Annals of Statistics*, vol. 29, no. 5, pp. 1189–1232, 2001. [Online]. Available: <https://doi.org/10.1214/aos/1013203451>